Predicting 30-day Readmission Risk Among Patients with Diabetes

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OBJECTIVES

Our objective was to build a model that predicts 30-day readmission among diabetic patients using clinical, demographic, medication, and hospitalization data.

Predicting 30-day Readmission Risk Among Patients with Diabetes

A summary of our exploratory data analysis

1. 30 day readmission rate

This highlights that the dataset is imbalanced, with far fewer 30-day readmissions.

2. Age distribution

This helps identify which age groups are most represented in the dataset (left skewed indicates mostly older patients.

3. Readmission by age

Helps reveal whether older or younger patients had higher readmission risk. Data does not reveal any meaningful conclusion.

4. Time in hospital

Describes the typical length of hospitalization in the dataset. Right skewed data reveals shorter time period.

5. Time in hospital *by 30-day readmission*

Helps assess whether longer stays are associated with readmission. Absolutely no indication of that!

6. Number of Medications

Patients who were readmitted had a slightly higher median number of medications than patients who were not readmitted.

7. Readmission by A1C Result

Shows the proportion of readmitted and non-readmitted patients within each A1C (blood sugar) result category.

8. Medication Change

Those with medication changes had a *slightly* higher readmission rate.

Diabetes 130-US Hospitals for Years 1999-2008 - UCI Machine Learning Repository

Data from UC Irvine ML Repository

Dataset Overview

1

101,766 encounters

2

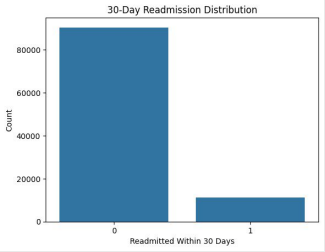
50 variables

3

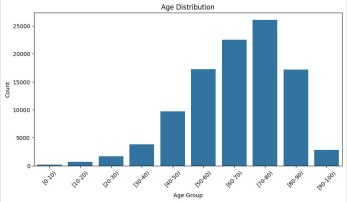
3 readmission categories: no readmission, readmission > 30 days, and readmission within 30 days.

EDA Visuals

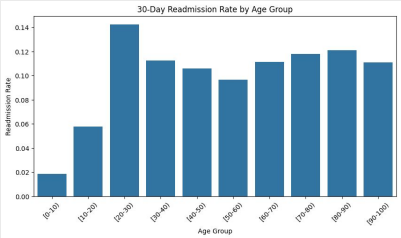
Readmission + Age



30-day readmission distribution

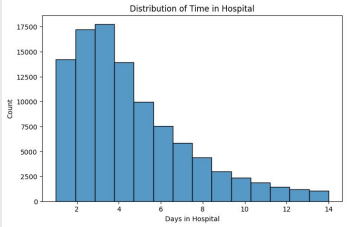


Age distribution

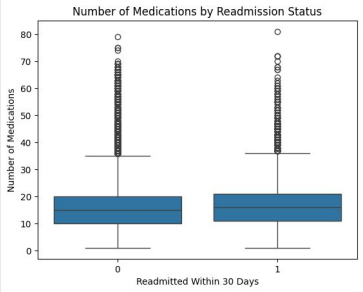


30-day readmission by age group

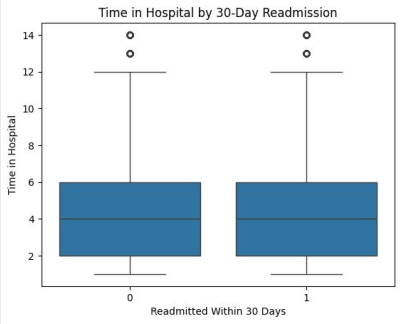
Readmission + Time in Hospital



Distribution of Time in Hospital



# of Meds by readmission status



Time in hospital by 30-day readmission

Key Conclusions

1

30-day readmission was uncommon

Most patients were not readmitted within 30 days, showing a strong class imbalance.

~80k

(wasn't readmitted)

2

No single EDA variable clearly explained readmission

The outcome likely depends on multiple factors, supporting the need for machine learning models.

3

Age was not a strong standalone predictor

Older patients were more common in the dataset, but readmission rates did not show a clear age-based pattern.

4

Length of hospital stay showed little difference

Patients who were readmitted and not readmitted had *very* similar hospital stay distributions.

5

Medication burden may matter

Readmitted patients had a *slightly* higher median number of medications, suggesting medication complexity may relate to readmission risk.

BOTTOM LINE:

30-day readmission is complex and cannot be predicted well from one variable alone.

Feature Engineering

Age Transformation

The original age variable was stored as ranges, such as [50–60) or [70–80). We converted each range into a numeric midpoint, allowing the models to treat age as a continuous numeric feature.

Diagnosis Grouping

The original diagnosis columns used many detailed medical codes. We grouped these into broader categories such as circulatory, respiratory, digestive, diabetes, injury, genitourinary, musculoskeletal, neoplasms, and other.

Medication-Based Features

We created two new medication features:

- Number of medication changes: counts how many diabetes medications were increased or decreased.
- Number of medications used: counts how many diabetes medications were active for each patient.

Why Feature Engineering Was Needed

The original dataset included several variables that were difficult for models to interpret directly, such as age ranges, diagnosis codes, and many individual medication columns. We transformed these into cleaner, more meaningful features.

Model Prep

Before training the models, we prepared the dataset so it could be used properly by machine learning algorithms. This step was important because the dataset included a mix of numeric variables, categorical variables, missing values, and an imbalanced target variable.

Train-Test Split

We split the dataset into two parts:

Training set: 80%
Used to teach the models patterns in the data.

Testing set: 20%
Used to evaluate how well the models performed on new, unseen data.

Scaling numeric features

Numeric features were scaled so that variables measured on different ranges would not unfairly influence the models. For example, variables like number of medications, time in hospital, and number of procedures may have different ranges. Scaling helps models treat these features more consistently.

Encoding categorical features

Categorical variables were converted into numeric format using one-hot encoding.

This was necessary because machine learning models cannot directly interpret text categories like race, gender, admission type, diagnosis category, or medication status.

HANDLING MISSING VALUES

NUMERIC

Missing values were filled using the median.

CATEGORICAL

Missing values were filled using the most frequent category.

→ **Imputation!**

Logistic Regression Model

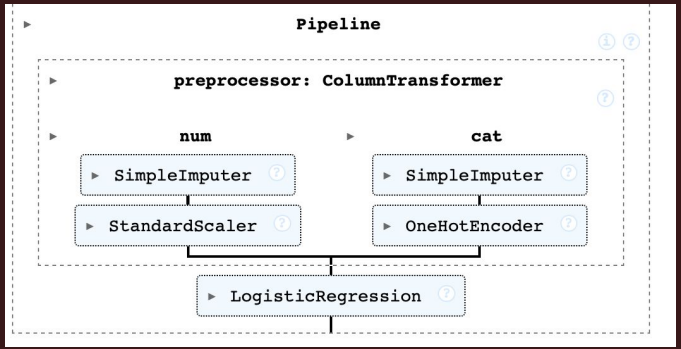
Why Logistic Regression?

Logistic Regression is a common classification model used to estimate the probability of a patient being readmitted within 30 days.

For our Project:

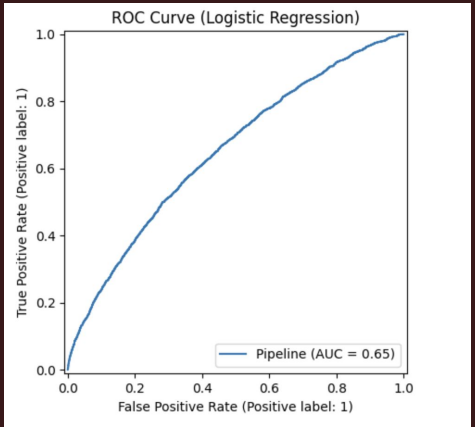
Because our target variable is binary (readmitted or not), logistic regression was an appropriate first model.

Pipeline Structure



The pipeline automatically handled preprocessing before training the Logistic Regression model. Numeric variables were imputed and scaled, while categorical variables were one-hot encoded.

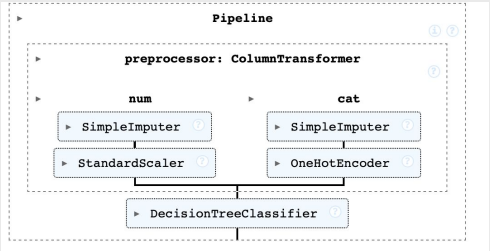
ROC Curve



The ROC-AUC score of 0.65 suggests the model performed better than random guessing, but still had only moderate ability to distinguish between patients who were readmitted and those who were not.

Tree-Based Models

Decision-Tree

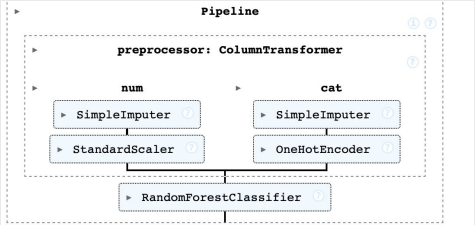


Pipeline

```
              precision    recall  f1-score   support

           0       0.92      0.69      0.79     18082
           1       0.18      0.54      0.27      2271

    accuracy                           0.68     20353
   macro avg       0.55      0.62      0.53     20353
weighted avg       0.84      0.68      0.73     20353

ROC-AUC: 0.6582987253478223
```

The Decision Tree model captured more complex relationships between patient variables, but overall predictive performance remained moderate with an ROC-AUC of about 0.66.

Random Forest



Pipeline

```
              precision    recall  f1-score   support

           0       0.92      0.70      0.80     18082
           1       0.18      0.53      0.27      2271

    accuracy                           0.68     20353
   macro avg       0.55      0.61      0.53     20353
weighted avg       0.84      0.68      0.74     20353

ROC-AUC: 0.6653428622122683
```

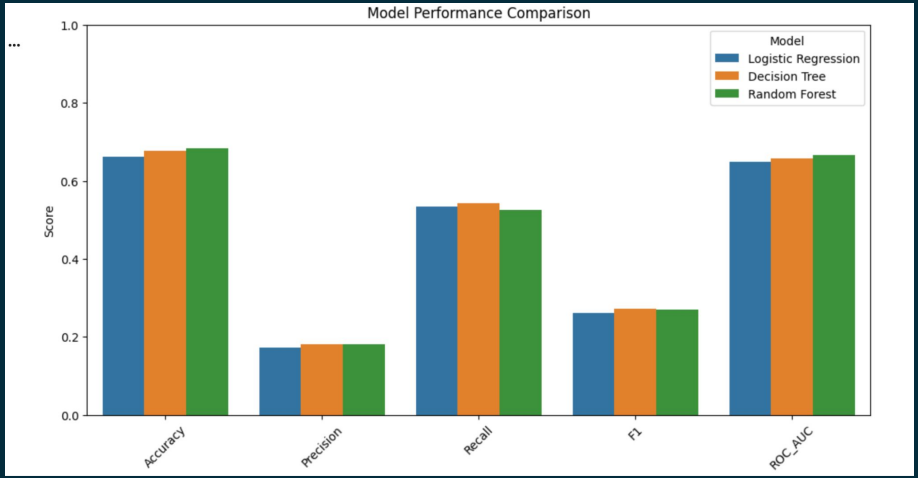
Random Forest produced performance similar to the Decision Tree model, suggesting that combining multiple trees only provided a modest improvement for this dataset.

Model Comparison

1

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
0	Logistic Regression	0.661033	0.172052	0.534566	0.260320	0.649333
1	Decision Tree	0.676706	0.181899	0.542492	0.272446	0.658299
2	Random Forest	0.682749	0.181721	0.526200	0.270148	0.665343

2



The three models produced relatively similar results across all evaluation metrics, with Random Forest achieving the highest ROC-AUC score, although the improvement over the other models was modest. Overall, no single model dramatically outperformed the others, suggesting that predicting 30-day hospital readmission is a complex problem influenced by many interacting patient, medication, and hospitalization factors.

Thank you for a great semester!!